

VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

SMART HOSTEL COMPLAINT MANAGEMENT AND REAL-TIME RESOLUTION SYSTEM

Submitted By:

RITHISH K

24MIC0145

M.Tech. Computer Science and Engineering

Submitted to :

Dr. ARUN PRASATH G

Assistant Professor Sr. Grade 2

Department of Information Security

School of Computer Science and Engineering

VIT , Vellore

For

CSI3023- Advanced Server Side Programming

School of Computer Science and Engineering

October 2026

ABSTRACT

Hostel complaint management is an important administrative process that requires efficient complaint registration, staff allocation, progress monitoring, and timely resolution. Traditional complaint-handling approaches may involve manual assignment, limited status visibility, and delays in identifying unresolved complaints. These limitations can affect coordination between students, maintenance staff, and hostel administrators.

The Smart Hostel Complaint Management and Real-Time Resolution System is a web-based application designed to centralize complaint handling and improve the coordination of complaint resolution. The system is developed using Java 21 and Spring Boot for backend processing, MySQL for data storage, and HTML, CSS, and JavaScript for the user interface. It incorporates automated staff assignment based on staff specialization, availability, and workload, along with severity-based Service Level Agreement (SLA) monitoring and escalation of overdue complaints. WebSocket-based communication supports real-time status updates, while role-based access control separates the responsibilities of students, staff, wardens, and administrators.

The system also supports complaint progress tracking and student verification of resolved complaints, with an option to reopen a complaint when necessary. The proposed solution aims to improve transparency, accountability, and coordination in hostel complaint management. Its effectiveness is to be evaluated through functional testing of complaint submission, staff assignment, SLA monitoring, escalation, status updates, and access control.

Keywords: Hostel Complaint Management, Spring Boot, Automated Staff Assignment, Service Level Agreement (SLA), Real-Time Notifications, WebSocket, Role-Based Access Control, Complaint Resolution.

1. INTRODUCTION

1.1 Background

Hostel facilities accommodate a large number of students and require effective maintenance and administrative support. Students may encounter issues related to electrical systems, plumbing, sanitation, room maintenance, and other hostel facilities. Addressing these complaints promptly is essential for maintaining a safe and comfortable living environment.

Traditional complaint management methods, such as manual registers, verbal communication, and phone calls, can make it difficult to track complaint progress, allocate staff efficiently, and monitor resolution deadlines. A centralized digital system can improve information accessibility, accountability, and coordination among students and hostel administrators.

1.2 Motivation

The motivation for this project is to improve the efficiency and transparency of hostel complaint handling through automation and real-time communication. Manual assignment and limited progress visibility may lead to delays and difficulties in identifying unresolved complaints.

The proposed system aims to simplify complaint registration, assist staff allocation, monitor resolution deadlines, and provide timely status updates. It also enables students to verify complaint resolution, helping ensure that complaints are addressed appropriately.

1.3 Project Overview

The Smart Hostel Complaint Management and Real-Time Resolution System is a web-based application developed using Java 21, Spring Boot, MySQL, HTML, CSS, and JavaScript. It provides separate access for students, maintenance staff, wardens, and administrators.

The system supports centralized complaint registration and tracking, automated staff assignment based on specialization, availability, and workload, and severity-based Service Level Agreement (SLA) monitoring. It also incorporates escalation of overdue complaints, WebSocket-based real-time updates, and role-based access control.

The application is designed to improve coordination between stakeholders and provide a structured workflow for complaint submission, assignment, resolution, and verification.

2. PROBLEM STATEMENT

Traditional hostel complaint management often relies on manual registers, verbal communication, or basic digital forms. These approaches may lack centralized tracking, systematic staff allocation, and effective monitoring of complaint resolution deadlines.

The absence of automated staff assignment based on specialization, availability, and workload can lead to inefficient resource utilization. Limited real-time status visibility makes it difficult for students and hostel administrators to monitor complaint progress. Furthermore, without severity-based deadlines and automated escalation, overdue complaints may remain unresolved without timely attention.

Therefore, there is a need for a centralized complaint management system that automates staff assignment, monitors resolution deadlines, provides real-time status updates, and supports transparent complaint verification and escalation.

3. OBJECTIVES

The primary objective of the Smart Hostel Complaint Management and Real-Time Resolution System is to develop a centralized web application that improves the management and resolution of hostel complaints.

The specific objectives are:

1. **Centralized Complaint Management:** Provide a unified platform for submitting, tracking, and managing hostel complaints.
2. **Automated Staff Assignment:** Allocate complaints based on staff specialization, availability, and current workload.
3. **Severity-Based SLA Monitoring:** Define resolution deadlines according to complaint severity and monitor overdue complaints.
4. **Automated Escalation:** Identify overdue complaints and notify responsible personnel for further action.
5. **Real-Time Communication:** Provide complaint status updates through WebSocket-based communication.
6. **Resolution Verification:** Allow students to verify resolved complaints or reopen them when necessary.
7. **Role-Based Security:** Restrict system access according to the roles and responsibilities of students, staff, wardens, and administrators.

6. LITERATURE REVIEW

6.1 Review of Existing Research

Research on digital complaint management, hostel administration, IT service management, task allocation, and real-time web applications provides the foundation for developing a centralized hostel complaint management system. Existing approaches demonstrate the importance of digitizing complaint registration, maintaining complaint records, tracking resolution progress, and improving communication between users and administrators. Web-based systems can make complaint information easier to access and help organize the complaint-handling process.

Studies on IT Service Management (ITSM) provide useful principles for managing service requests through defined processes, accountability, and service-level objectives. Iden and Eikebrokk (2013) conducted a systematic literature review of ITSM and IT Infrastructure Library (ITIL) implementation research, identifying implementation motivations, critical success factors, implementation status, and benefits as major research themes. Serrano et al. (2021) reviewed ITSM literature to examine its benefits, challenges, opportunities, and implementation practices. These studies provide a conceptual basis for designing a structured complaint-resolution workflow.

Research on complaint and hostel management is relevant to the development of online complaint registration, centralized storage, role-based access, and progress tracking. Such systems provide a digital alternative to complaint registers, telephone calls, and informal communication. However, a system that only records complaints may not provide sufficient support for selecting suitable staff, monitoring resolution deadlines, or escalating overdue requests. These functions are particularly important when maintenance teams must handle complaints of different categories and urgency levels.

Task-allocation research provides another relevant perspective. Allocation approaches can consider the suitability of an available resource for a task, along with workload distribution. In the proposed project, these principles are applied through rule-based staff assignment using specialization, availability, and current workload. This should not be confused with machine learning or artificial intelligence: the project uses defined assignment criteria rather than a predictive model.Real-time web application research is also relevant because complaint resolution involves several participants who need timely status information. WebSocket-based communication can support updates between the server and connected clients without requiring every update to be obtained through repeated manual refreshes. However, real-time communication alone does not ensure that complaints are assigned correctly, resolved within their deadlines, or verified by students. It must be integrated with the complaint lifecycle and administrative workflow.

Based on these research areas, the present project brings together complaint registration, rule-based staff assignment, severity-based Service Level Agreement (SLA) monitoring, overdue complaint escalation, real-time status updates, role-based access, and student verification of resolution. The research gap is therefore framed as an integration and workflow gap: the need to combine these operational functions in a single hostel-focused application, rather than claiming that no previous system has implemented any of them individually.

6.2 Comparative Analysis of Research Papers

The following table organizes 15 relevant papers and research areas by title, authors, domain, methodology, limitations, and relevance to the proposed system. For the hostel-specific entries, the bibliographic details must be checked against the original publication before these are treated as verified references.

S. No.	Paper title and authors	Domain related to project	Key methodology / approach	Limitations or scope boundaries	Research gap relevant to this project
1	Design and Development of Online Hostel Management System to Improve Application and Management Process — Wong and Mahdin (2022)	Hostel management	Web-based hostel management using database-backed application functions	General hostel management; verify the paper's exact complaint-handling features	Integrating automated assignment and SLA escalation into hostel complaint handling

2	Web-based Campus Complaint Management System (WCCMS) — Alifat et al. (2022)	Campus grievance management	Web-based complaint submission and management	Verify the exact assignment and tracking functions described in the full paper	Linking complaint tracking with staff workload and resolution deadlines
3	Hostel Management Systems for University Selangor — Ahmad Fahmi, Sharip, and Muhammad	Hostel maintenance and administration	Digital maintenance and complaint-management workflow	The full publication details and exact methodology require verification	Combining maintenance tracking with automated staff allocation
4	A Literature Review of a Web-Based Complaint Management System — Nandini et al. (2025)	Institutional complaint management	Literature review of web-based complaint systems	A review does not itself validate a complete integrated implementation	Identifying and implementing a unified complaint lifecycle
5	StudentConnect: A Digital Solution for Managing Late Permissions and Maintenance Complaints in Hostels — Gnanagurusubbiah, Manivannan, and Kumar (2026)	Hostel administration	Digital permission and maintenance complaint workflows	Verify publication status and whether workload-aware assignment is included	Integrating staff suitability, SLA monitoring, and escalation
6	StaySmart: A Web-Based Hostel Management System for Student Accommodation — Yusof et al. (2026)	Hostel management	Web-based accommodation management with complaint-related functions	Exact implementation details and publication metadata require verification	Coordinating complaint categorization, staff assignment, and resolution verification
7	Web Based Institute Complaint Management System — Bakode et al. (2026)	Institutional maintenance	Online complaint reporting and worker coordination	Verify the extent of automation and testing in the original paper	Combining assignment history with severity-based escalation

8	E-Hostel Management System for Fair Room Allocation and Complaint Resolution Tracking — Ydv et al. (2026)	Hostel administration	Digital hostel processes and complaint-resolution tracking	The reported scope may emphasize general hostel management	Providing an integrated assignment and overdue-monitoring workflow
9	Automation Tool for Grievances Using Artificial Intelligence — Procedia Computer Science (2026)	Automated grievance handling	Reported AI-assisted categorization and prioritization	Different technical approach; publication details and evaluation should be verified	A simpler, explainable rule-based alternative for staff assignment and SLA handling
10	Implementing IT Service Management: A Systematic Literature Review — Iden and Eikebrokk (2013)	IT service and incident management	Systematic literature review of ITSM and ITIL implementation	Broad ITSM focus rather than hostel maintenance	Applying structured service-management principles to hostel complaints
11	An IT Service Management Literature Review: Challenges, Benefits, Opportunities and Implementation Practices — Serrano et al. (2021)	IT service management	Systematic literature review of 47 selected articles	General ITSM review; not an evaluation of a hostel application	Adapting service-management practices to a student-facing complaint workflow
12	Key Factors in Achieving Service Level Agreements (SLA) for Information Technology (IT) Incident Resolution — Swain and Garza (2023)	SLA and incident resolution	Empirical analysis of factors associated with SLA achievement	IT incident data may not represent hostel maintenance operations	Applying severity-based deadlines and overdue monitoring in hostel complaints
13	A Survey on Intelligent Management of Alerts and Incidents in IT Services —	Incident and alert management	Survey of approaches to incident and alert handling	General IT service context; not specifically hostel complaints	Translating incident identification and escalation concepts into

	publication details to verify				maintenance workflows
14	Competency Adjustment and Workload Balancing Framework in Multirobot Task Allocation — Lee et al. (2018)	Task allocation and workload balancing	Competency-aware task assignment and workload balancing	Multi-robot assumptions differ from human staff scheduling	Using staff specialization and current workload as assignment criteria
15	Protocols, Reactive Architectures, and Computing Platforms for Low-Latency, High-Concurrency Web Applications: A Systematic Literature Review — publication details to verify	Real-time web applications	Review of communicatio n protocols and application architectures	General web architecture; does not evaluate hostel complaint resolution	Integrating real-time updates with assignment, SLA monitoring, and verification

6.3 Identified Research Gaps

The comparative review identifies the following gaps relevant to the proposed system:

1. Fragmented complaint management: Complaint registration, staff assignment, status tracking, and resolution verification may be handled by separate processes rather than one integrated workflow.

2. Limited workload-aware assignment: A complaint-management workflow can benefit from considering staff specialization, availability, and current workload when assigning maintenance tasks.

3. Insufficient deadline monitoring: Complaint systems need a structured approach to setting resolution deadlines according to severity and identifying overdue complaints.

4. Limited automated escalation: Identifying overdue complaints is more useful when the system also initiates an escalation or notification to the responsible personnel.

5. Lack of integrated real-time updates: Live communication must be connected to complaint status changes so that relevant users can receive timely information.

6. Resolution verification: A structured mechanism for students to verify a resolution or reopen a complaint can improve accountability after staff mark a complaint as resolved.

7. Need for an integrated, explainable workflow: The reviewed domains suggest an opportunity to combine these functions in one application using transparent, rule-based logic instead of requiring a complex predictive assignment model.

6.4 Research Gap Addressed by the Proposed System

The proposed Smart Hostel Complaint Management and Real-Time Resolution System addresses the identified integration gap by combining centralized complaint management, rule-based staff assignment, severity-based SLA monitoring, overdue complaint escalation, WebSocket-based status updates, and student resolution verification.

Unlike an approach that focuses only on recording complaints, the proposed application is designed to support the complaint lifecycle from submission to verification. Staff assignment considers specialization, availability, and workload, while deadline monitoring and escalation support timely administrative attention. Role-based access separates the functions available to students, staff, wardens, and administrators.

The project uses Java 21, Spring Boot, MySQL, HTML, CSS, JavaScript, and WebSocket-based communication. Its contribution is the implementation of these functions within a single hostel-focused application. The effectiveness of the implementation must be established through functional testing and documented outputs; comparative performance improvements should not be claimed without supporting measurements.

7. EXISTING SYSTEM

7.1 Overview

In traditional hostel environments, students report maintenance issues such as electrical faults, plumbing problems, sanitation concerns, and room maintenance requirements through complaint registers, telephone calls, verbal communication, or online forms. Hostel staff or administrators record these complaints and coordinate with maintenance personnel to resolve them.

Some digital systems provide online complaint registration, centralized storage, and basic status tracking. These features reduce paperwork and improve access to complaint records. However, the functionality available varies between systems, and basic complaint tracking alone may not provide a complete mechanism for assignment, deadline monitoring, escalation, and resolution verification.

7.2 Limitations of the Existing System

1. **Manual Complaint Handling:** Paper-based reporting and informal communication can make complaint records difficult to organize and retrieve.
2. **Inefficient Staff Allocation:** Without a systematic assignment mechanism, selecting staff based on specialization, availability, and workload may require manual coordination.
3. **Limited Progress Visibility:** Students may need to contact hostel authorities to obtain updates when complaint status information is not readily available.
4. **Lack of Structured SLA Monitoring:** Some basic systems may not define severity-based resolution deadlines or automatically identify overdue complaints.
5. **Delayed Escalation:** Complaints that remain unresolved may require manual follow-up by students or administrators.
6. **Limited Resolution Verification:** Systems without a student confirmation mechanism may not provide a clear final verification step after staff report a complaint as resolved.

7.3 Summary

Existing systems demonstrate the value of digitizing hostel complaint registration and record management. However, the extent of automation, monitoring, and real-time communication depends on the particular implementation. The proposed project addresses these operational requirements through a unified workflow incorporating rule-based staff assignment, severity-based SLA monitoring, escalation, real-time updates, and student resolution verification.

8. PROPOSED SYSTEM

8.1 Overview

The Smart Hostel Complaint Management and Real-Time Resolution System is a web-based application designed to centralize hostel complaint registration, staff assignment, progress monitoring, and resolution verification. It provides separate access for students, maintenance staff, wardens, and administrators to support a structured complaint-resolution process.

The system uses Java 21 and Spring Boot for backend processing, MySQL for data storage, and HTML, CSS, and JavaScript for the user interface. WebSocket-based communication supports real-time complaint status updates.

8.2 Key Features

1. **Centralized Complaint Management:** Enables students to submit complaints and track their progress through a unified application.
2. **Automated Staff Assignment:** Selects suitable staff based on specialization, availability, and current workload.
3. **Severity-Based SLA Monitoring:** Associates complaint severity with configured resolution deadlines and monitors overdue complaints.
4. **Automated Escalation:** Identifies overdue complaints and initiates escalation notifications for administrative attention.
5. **Real-Time Status Updates:** Uses WebSocket-based communication to deliver complaint updates to connected users.
6. **Resolution Verification:** Allows students to verify a resolved complaint or reopen it when the issue remains unresolved.
7. **Role-Based Access Control:** Separates system functions and permissions for students, staff, wardens, and administrators.
8. **Assignment Management:** Maintains complaint assignment information to support staff coordination and assignment tracking.

8.3 Novelty and Contribution

The main contribution of the proposed system is the integration of complaint management, rule-based staff assignment, SLA monitoring, escalation, real-time updates, and student verification within a single hostel-focused application.

Rather than relying solely on complaint registration and manual follow-up, the system provides a structured workflow from complaint submission to resolution verification. Staff selection follows defined criteria, making the assignment process systematic and explainable.

8.4 Expected Benefits

- Improved organization of hostel complaint records.
- More systematic staff allocation.
- Better visibility into complaint status and deadlines.
- Timely identification of overdue complaints.
- Improved coordination among students, staff, wardens, and administrators.
- Greater transparency in the complaint-resolution process.

9. METHODOLOGY

9.1 Overview

The system follows a structured workflow to manage hostel complaints from submission to resolution verification. It combines complaint validation, rule-based staff assignment, severity-based Service Level Agreement (SLA) monitoring, overdue complaint escalation, and real-time status communication.

9.2 Complaint Management Workflow

Step 1: Complaint Submission

The student submits a complaint by providing the required details, including the title, description, category, and severity.

Step 2: Validation and Storage

The backend validates the submitted information and stores the complaint in the MySQL database.

Step 3: Priority Assessment

The system uses the complaint's severity and configured rules to determine its resolution deadline.

Step 4: Automatic Staff Assignment

The assignment logic evaluates staff specialization, availability, and current workload to identify a suitable staff member.

Step 5: Complaint Processing

The assigned staff member reviews the complaint, performs the required maintenance work, and updates the complaint status.

Step 6: SLA Monitoring and Escalation

A scheduled process checks complaint deadlines periodically. Overdue complaints are identified and escalated through the configured notification mechanism.

Step 7: Resolution Verification

After the complaint is marked as resolved, the student can verify the resolution or reopen the complaint if the issue persists.

9.3 Automatic Staff Assignment Algorithm

The assignment process uses predefined rules rather than a machine-learning model.

1. Retrieve the complaint details.
2. Identify staff members relevant to the complaint category or specialization.
3. Consider staff availability and current workload.
4. Select a suitable eligible staff member according to the implemented assignment rules.
5. Create the assignment record and update the relevant complaint information.
6. Handle existing assignments according to the system's assignment-validation rules.

9.4 SLA Monitoring and Escalation

The system associates complaint severity with configured resolution targets. A scheduled task checks complaint deadlines periodically and identifies overdue complaints. The escalation mechanism then initiates the configured notification or administrative follow-up process.

9.5 Real-Time Communication

WebSocket-based communication is used to support real-time complaint status updates. This helps connected users receive relevant changes without relying exclusively on manual page refreshes.

9.6 Implementation Process

The application is developed using a layered approach. The frontend provides role-specific interfaces, the Spring Boot backend processes requests and business logic, and MySQL stores complaint and assignment information. Security controls govern access to the system's role-specific functions.

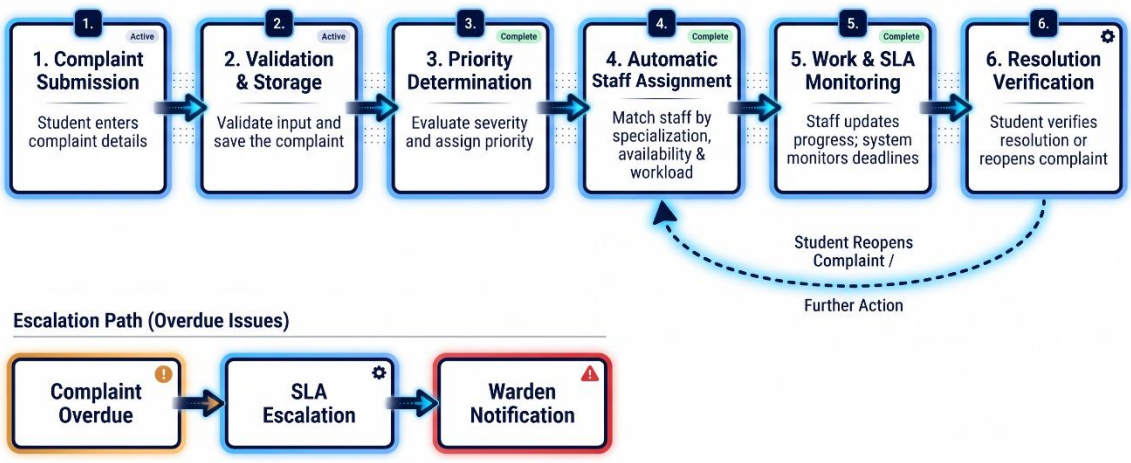


Figure 1:Methodology of the proposed hostel complaint management system.

10. SYSTEM DESIGN

10.1 System Architecture

The Smart Hostel Complaint Management and Real-Time Resolution System follows a layered web application architecture. The frontend provides user interfaces, the backend processes requests and business logic, the database stores application records, and the real-time communication component supports status updates.

The architecture consists of the following layers:

1. Presentation Layer

Provides role-specific interfaces for students, staff, wardens, and administrators using HTML, CSS, and JavaScript.

2. Application Layer

Uses Java 21 and Spring Boot to implement REST APIs, process user requests, execute business logic, and enforce access controls.

3. Processing and Real-Time Services

Handles automatic staff assignment, SLA monitoring, overdue complaint escalation, notifications, and WebSocket-based communication.

4. Data Layer

Uses MySQL to store complaint details, staff profiles, assignment records, and related application information.

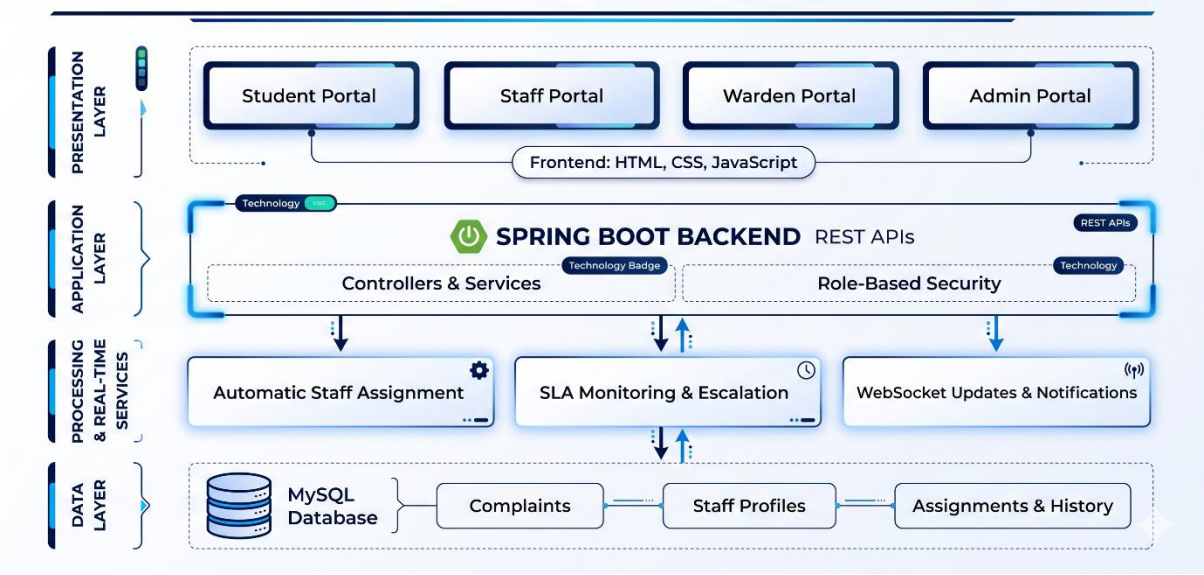


Figure 2:System Architecture Diagram

10.2 System Workflow

The workflow begins when a student submits a complaint. The backend validates the request and stores the complaint in the database. The assignment logic selects suitable staff according to the configured criteria. During complaint processing, the system monitors the resolution deadline and initiates escalation when applicable. After the complaint is marked as resolved, the student can verify the resolution or reopen the complaint.

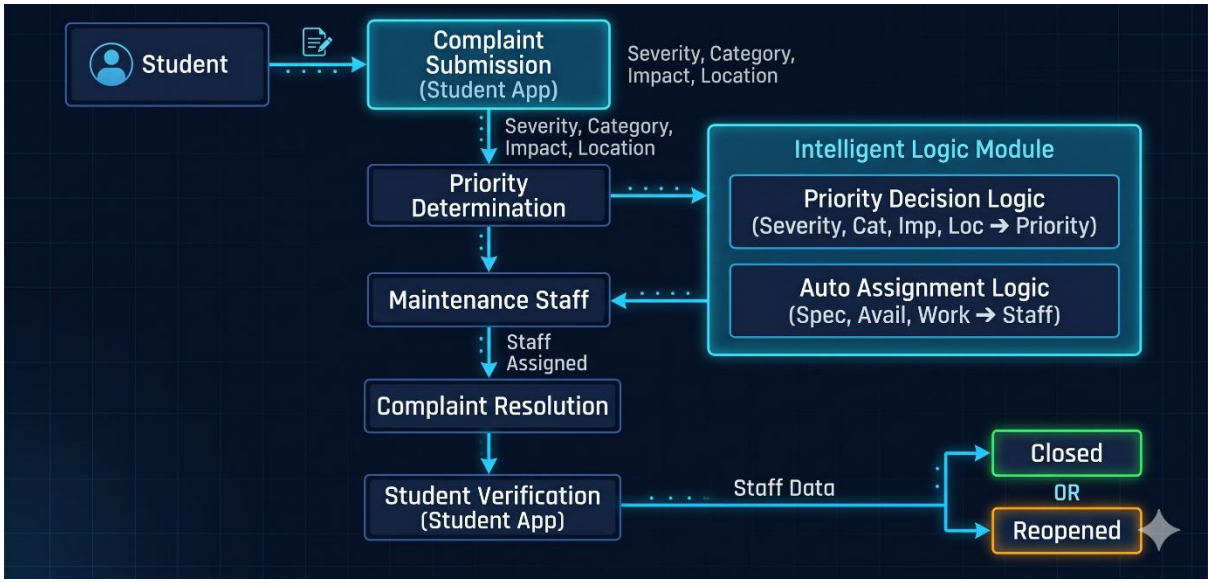


Figure 3: Complaint Workflow Flowchart

10.3 Use Case Design

The system includes four primary actors:

- **Student:** Submits complaints, views status, and verifies or reopens resolved complaints.
- **Staff:** Views assigned complaints and updates their progress.
- **Warden:** Monitors complaint handling and performs authorized administrative actions.
- **Administrator:** Manages system-level information and authorized user-related operations.

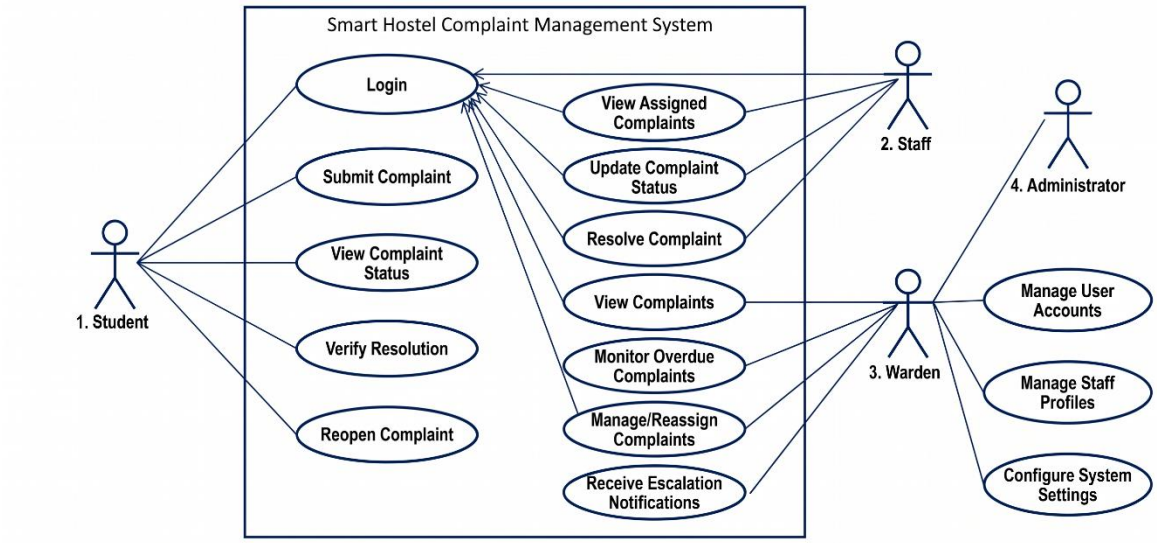


Figure 4:Use Case Diagram of the Proposed System

10.4 Database Design

The database stores the information required to manage complaints and staff assignments.

Complaint: Stores complaint details, category, severity, priority, and status.

Staff Profile: Stores staff details, specialization, availability, and current workload.

Assignment: Links complaints with staff members and records assignment information, including assignment date and status.

The assignment records support tracking the relationship between complaints and the staff responsible for handling them.

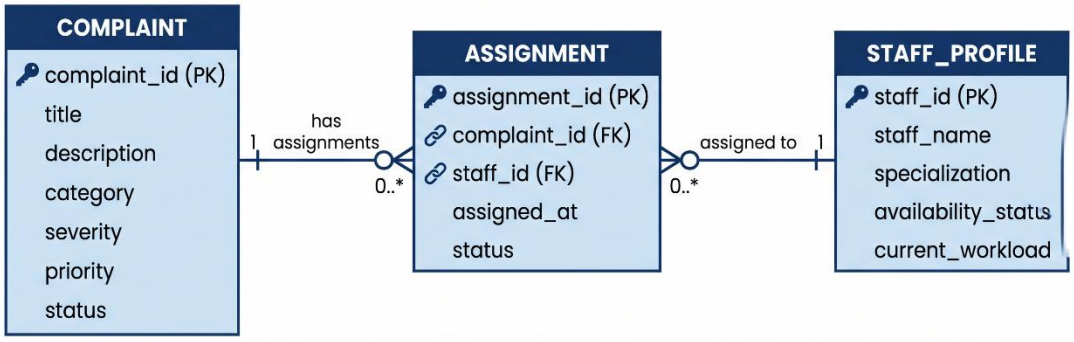


Figure 5:Entity–Relationship (ER) Diagram of the Proposed System.

10.5 Security Design

The system uses role-based access control to restrict operations according to user roles. Additional security measures include session management and Cross-Site Request Forgery (CSRF) protection, as configured in the application. These mechanisms help protect application operations and restrict unauthorized access.

11. TECHNOLOGIES USED

The Smart Hostel Complaint Management and Real-Time Resolution System is developed using a combination of frontend technologies, backend frameworks, database systems, and communication technologies. These tools support user interaction, complaint processing, data persistence, real-time updates, and application security.

11.1 Programming Languages and Frameworks

Java 21: Used to implement backend business logic, complaint processing, staff assignment, and SLA monitoring.

Spring Boot: Provides the backend application framework for developing REST APIs and organizing application components.

HTML5: Structures the web pages and user interfaces.

CSS3: Defines the layout, styling, and visual presentation of the application.

JavaScript: Supports client-side interactions and communication with backend endpoints.

11.2 Database and Data Access

MySQL: Stores complaint details, staff profiles, assignment records, and related application data.

Spring Data JPA and Hibernate: Support object-relational mapping and database access through Java entity classes and repositories.

11.3 Communication and Messaging

RESTful APIs: Enable communication between the frontend and backend for application operations.

WebSocket: Supports real-time communication and complaint status updates between the server and connected clients.

ActiveMQ Artemis: Provides message-broker capabilities for asynchronous messaging where configured and used by the application.

11.4 Security Technologies

Spring Security: Supports authentication and authorization mechanisms.

Role-Based Access Control (RBAC): Restricts application operations according to user roles.

CSRF Protection and Session Management: Help protect state-changing requests and manage authenticated user sessions.

11.5 Development Tools

Eclipse IDE: Used for application development, code editing, debugging, and project execution.

Maven: Manages project dependencies and the build process.

Git and GitHub: Support version control and source-code management.

Localhost: Used to run and test the application in the local development environment.

12. IMPLEMENTATION

12.1 Overview

The Smart Hostel Complaint Management and Real-Time Resolution System is implemented as a web-based application using Java 21, Spring Boot, and MySQL. The frontend is developed using HTML, CSS, and JavaScript. The application follows a layered architecture in which frontend requests are processed by backend controllers and services, while the database maintains complaint and assignment records.

12.2 Complaint Management Module

The complaint management module allows students to submit hostel-related complaints and view their complaint information. Complaint records contain details such as title, description, category, severity, priority, and status. The backend validates requests and stores complaint information in the MySQL database.

12.3 Automatic Staff Assignment Module

The assignment module allocates complaints to suitable maintenance staff using predefined rules. Staff specialization, availability, and current workload are considered during selection. Assignment records connect complaints with staff profiles and maintain assignment-related information.

12.4 SLA Monitoring and Escalation Module

The SLA monitoring module evaluates complaint deadlines according to configured severity levels. A scheduled process periodically checks complaints to identify overdue cases. When an overdue complaint is detected, the configured escalation mechanism initiates the relevant notification or administrative follow-up.

12.5 Real-Time Communication Module

The application uses WebSocket-based communication to support real-time complaint status updates. This enables connected clients to receive relevant updates as complaint information changes, reducing reliance on manual page refreshes.

12.6 Resolution Verification Module

After a complaint is marked as resolved, the student can verify the resolution or reopen the complaint if the issue persists. This provides a structured mechanism for confirming whether the reported problem has been addressed.

12.7 User Roles and Security Module

The application provides role-specific access for students, staff, wardens, and administrators. Access restrictions are intended to ensure that users can perform only the operations permitted for their roles. Session management and CSRF protection are used where configured in the application.

12.8 Database Integration

MySQL stores the application's persistent data. The principal entities discussed in the system design are Complaint, Staff Profile, and Assignment. These entities represent complaint details, staff information, and complaint-to-staff assignment records, respectively.

12.9 Development and Execution

The application is developed using Eclipse IDE and Maven-based project management. The backend runs in the local development environment, and the frontend communicates with the backend through the application's configured endpoints. Git and GitHub are used for source-code version control.

13. SCREENSHOTS OF IMPLEMENTATION AND OUTPUT

This section presents screenshots of the implemented Smart Hostel Complaint Management and Real-Time Resolution System. The screenshots demonstrate the application's user interface, complaint-handling workflow, staff assignment, status monitoring, and role-specific functionality.

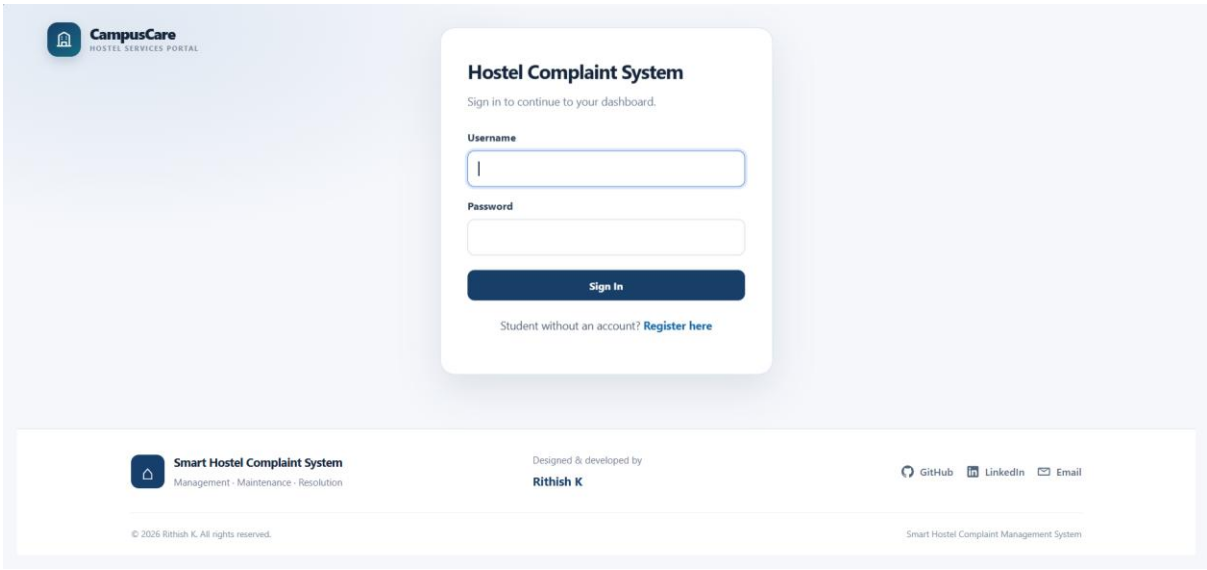


Figure 6: Login Page

The login page provides the entry point for authorized users to access the application according to their assigned roles.

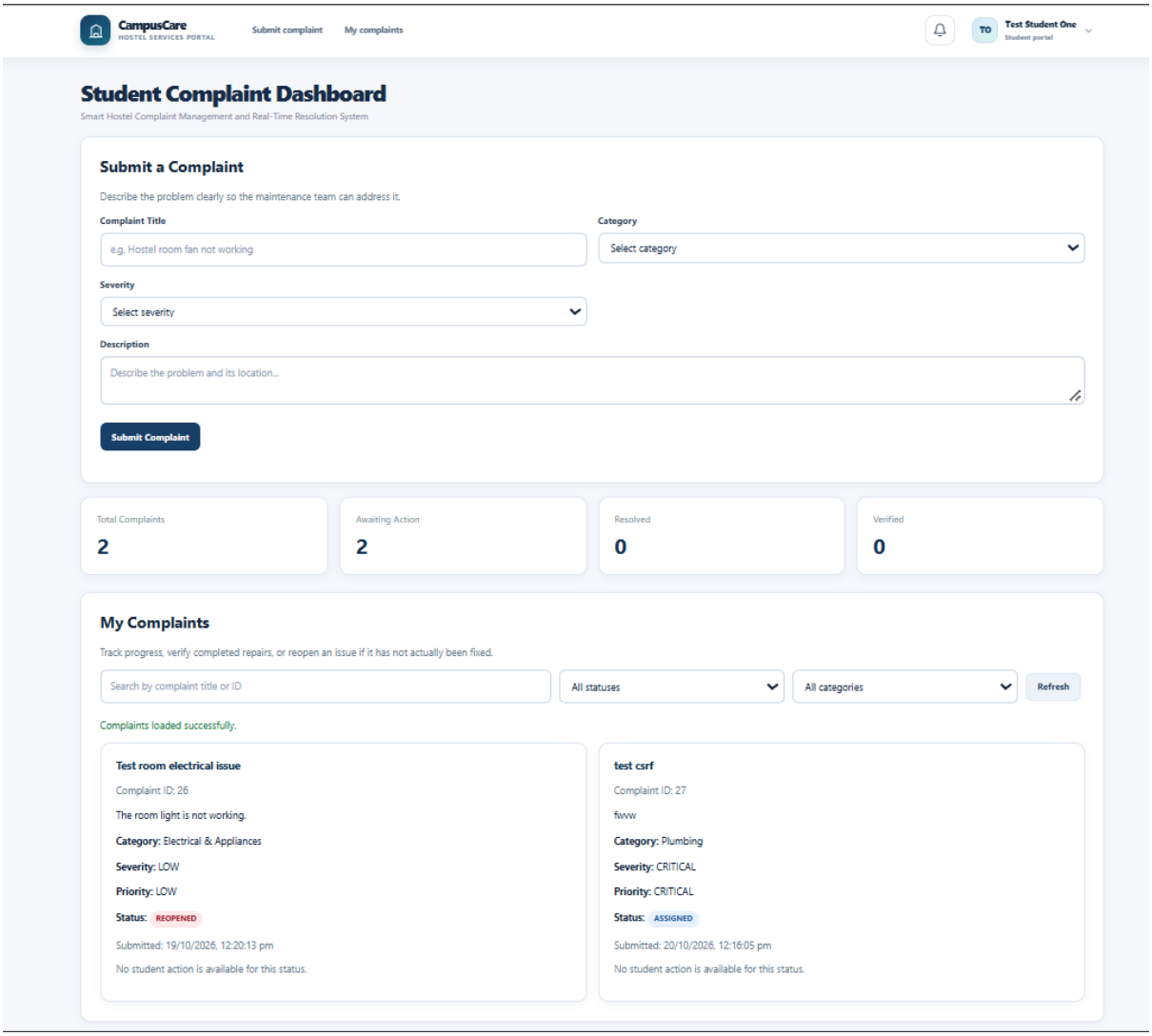


Figure 7: Student Dashboard

The student dashboard provides access to complaint-related functions, allowing students to submit complaints and monitor their progress.

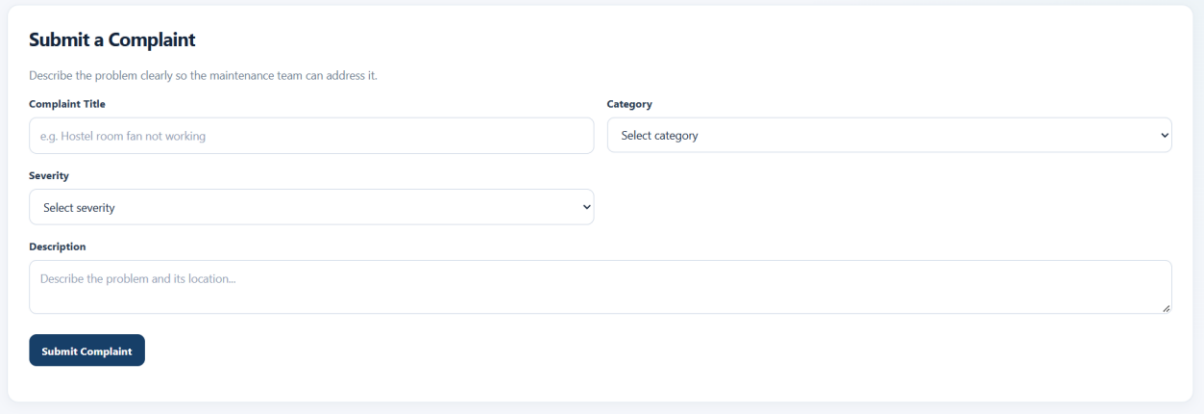


Figure 8:Complaint Submission Page

The complaint submission interface enables students to enter complaint details, including the title, description, category, and severity.

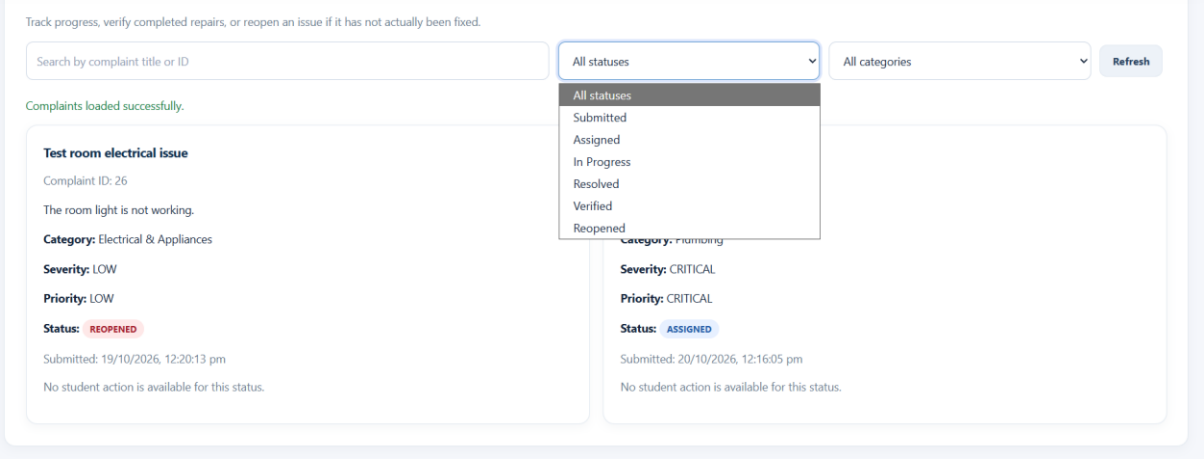


Figure 9: Complaint Tracking Page

The complaint tracking interface displays recorded complaints and their current statuses, helping students monitor progress.

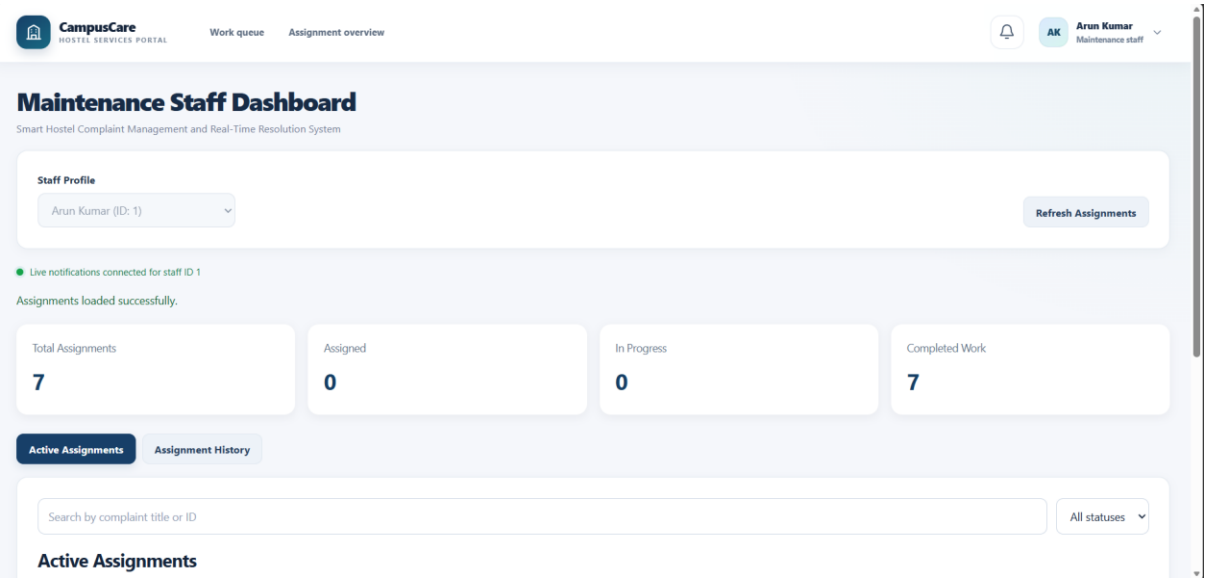


Figure 10: Staff Dashboard

The staff interface provides access to assigned complaints and supports complaint progress updates.

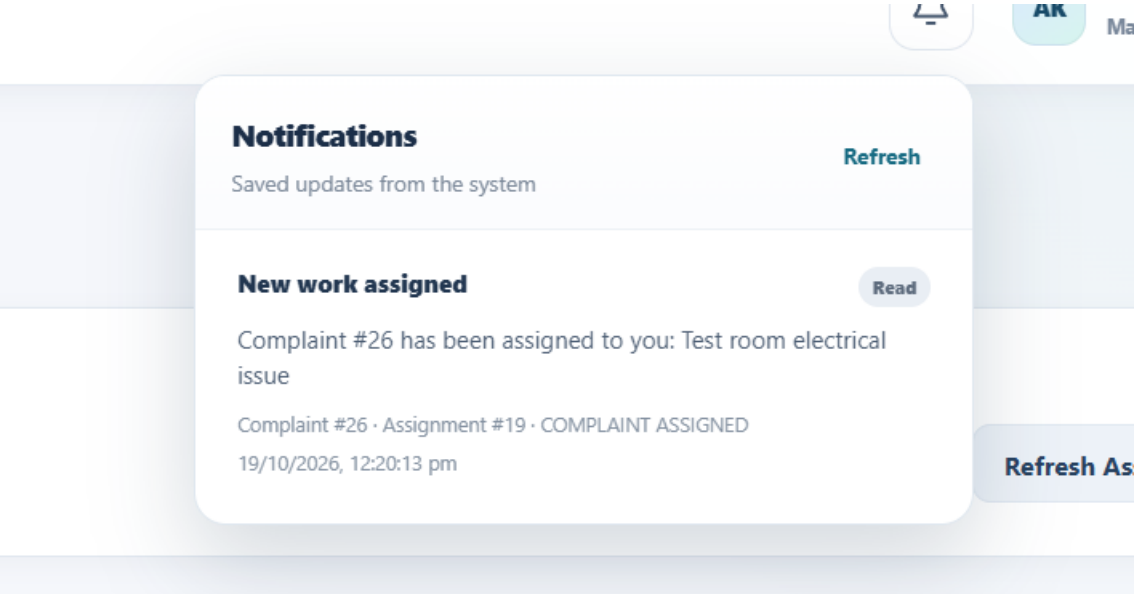


Figure 11: Automatic Staff Assignment

This screenshot demonstrates the assignment of a complaint to a suitable staff member according to the assignment rules implemented in the application.

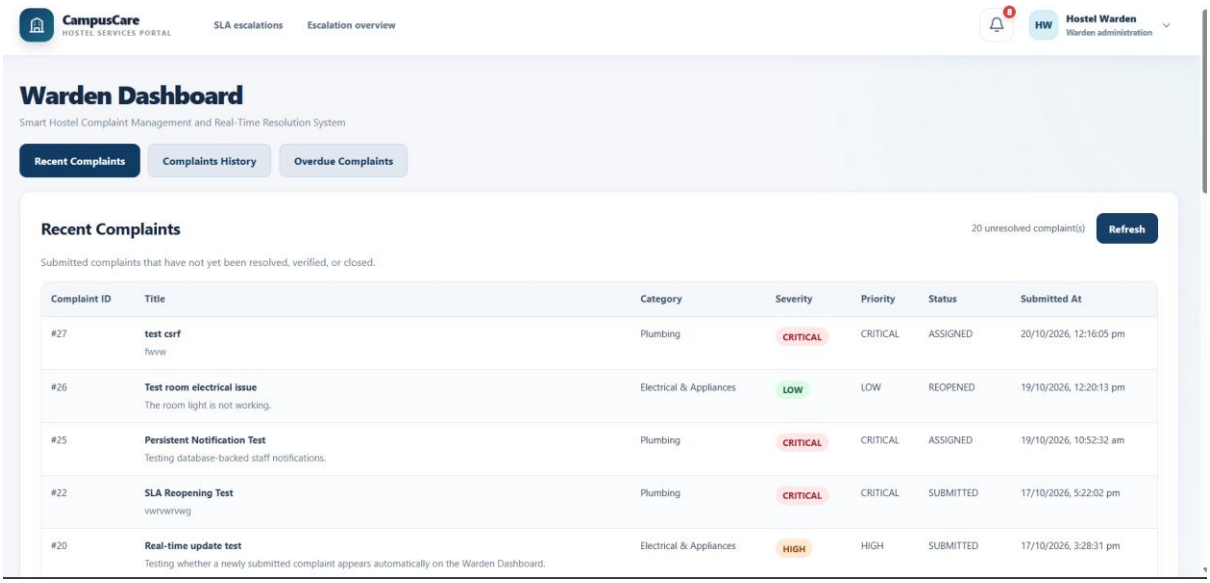


Figure 12: Warden Dashboard

The warden interface supports complaint monitoring and the administrative actions available to the warden role.

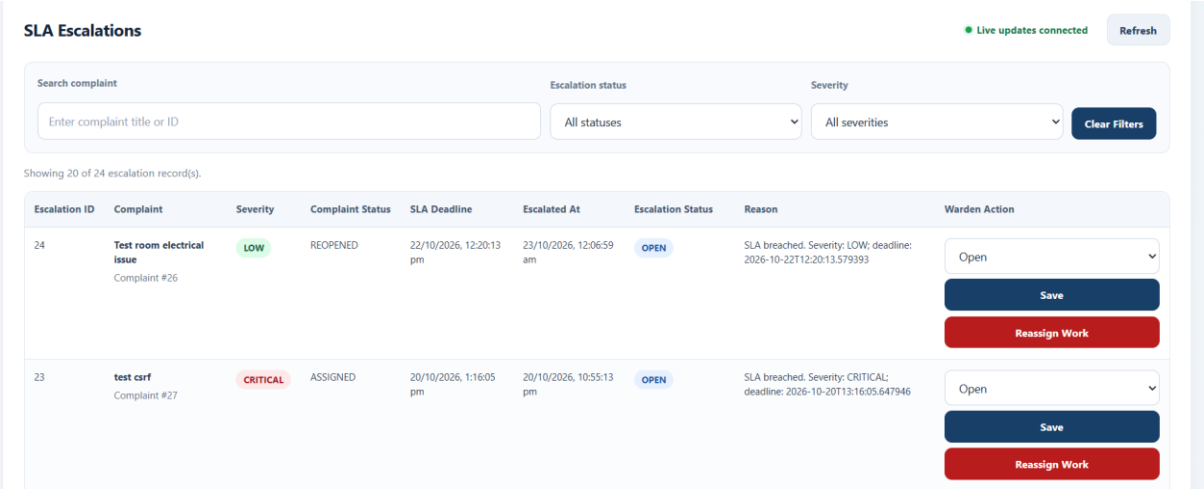


Figure 13: SLA Monitoring and Escalation

This screenshot demonstrates overdue complaint identification and the corresponding escalation or notification, where implemented.

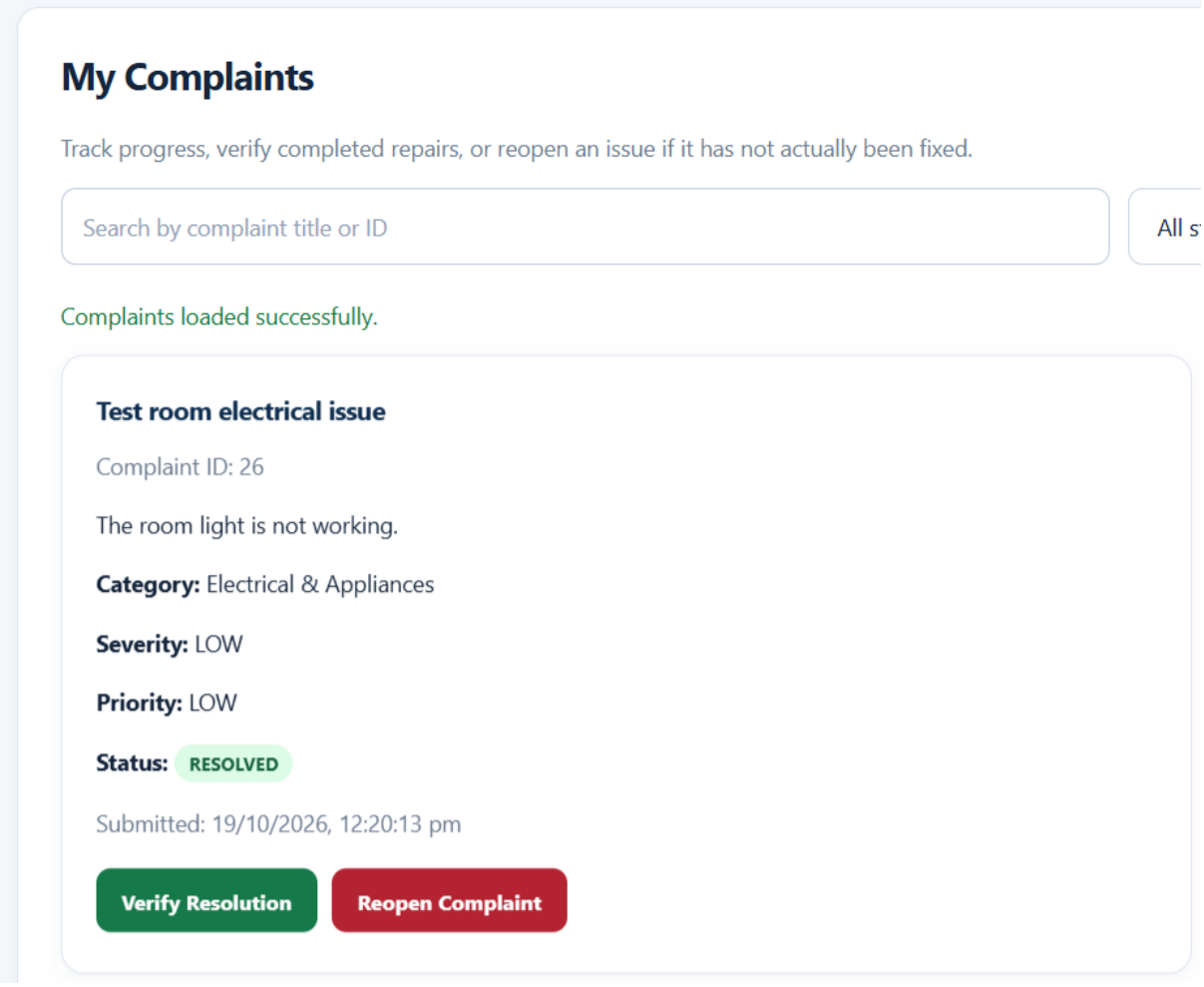


Figure 14: Resolution Verification

This screenshot demonstrates the student-facing resolution verification or complaint reopening workflow, where implemented.

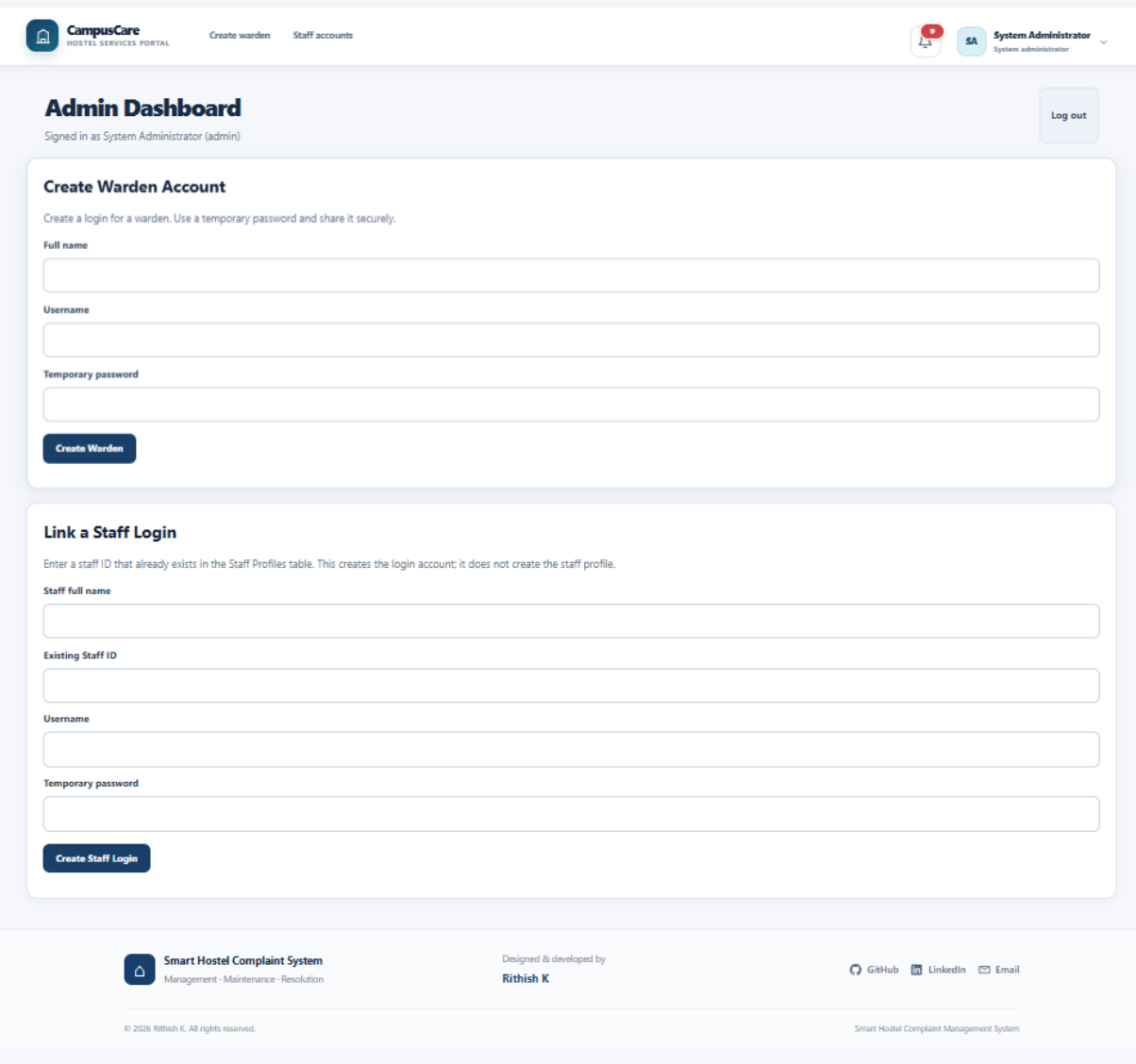


Figure 15: Administrator Interface

The administrator interface demonstrates the user-management and staff-management functions available in the implemented application.

14. RESULTS AND DISCUSSION

14.1 Overview

The Smart Hostel Complaint Management and Real-Time Resolution System is evaluated through functional testing of its major modules. The evaluation focuses on complaint submission, database storage, automatic staff assignment, SLA monitoring, escalation, real-time status updates, role-based access, and resolution verification.

The purpose of the evaluation is to determine whether the implemented functions operate according to the specified requirements. Test results are recorded based on the actual outputs observed during application execution.

14.2 Functional Testing

Test Case ID	Functionality Tested	Expected Result	Actual Result	Status
TC01	User Login	Authorized users can log in successfully.	Login functionality worked as expected.	PASS
TC02	Complaint Submission	Valid complaints are stored in the database.	Complaint submitted and stored successfully.	PASS
TC03	Complaint Retrieval	Recorded complaints are displayed correctly.	Complaint records were retrieved successfully.	PASS
TC04	Automatic Staff Assignment	Suitable staff are assigned according to configured rules.	Staff assignment operated according to the implemented criteria.	PASS
TC05	SLA Monitoring	Overdue complaints are identified.	SLA monitoring identified overdue complaints as expected.	PASS
TC06	Escalation	Configured escalation notifications are triggered.	Escalation functionality worked as expected.	PASS
TC07	Real-Time Updates	Connected clients receive complaint status updates.	Real-time status updates worked as expected.	PASS
TC08	Role-Based Access Control	Access is restricted according to user roles.	Role-based access restrictions worked as expected.	PASS
TC09	Resolution Verification	Students can verify resolution or reopen complaints.	Resolution verification and reopening worked as expected.	PASS

14.3 Discussion

The evaluation examines whether the system provides a structured complaint-resolution workflow. Complaint submission and retrieval tests assess the core complaint-management functionality. Assignment tests examine the implementation of staff-selection rules. SLA and escalation tests assess the identification and handling of overdue complaints.

Real-time communication tests evaluate whether complaint status changes are reflected for connected users. Security tests assess access restrictions across the supported roles.

Resolution verification tests examine whether students can confirm that reported problems have been addressed.

The final findings must be based on the recorded test outcomes. Any failed tests, implementation limitations, or incomplete functions should be documented and discussed. Quantitative performance claims should be included only when supported by measured results.

14.4 Summary

The functional testing covered nine major test cases across authentication, complaint management, staff assignment, SLA monitoring, escalation, real-time communication, access control, and resolution verification.

All nine test cases were recorded as passed based on the application's observed behavior during testing. The results indicate that the tested functionalities operated according to their intended requirements.

The testing confirms the functional behavior of the evaluated modules. Further quantitative performance testing would be required to establish response times, scalability, and behavior under concurrent workloads.

15. CONCLUSION

The Smart Hostel Complaint Management and Real-Time Resolution System was developed to improve the organization, tracking, and resolution of hostel complaints through a centralized web-based application. The system integrates Java 21, Spring Boot, MySQL, and web technologies to support complaint processing and coordination among students, maintenance staff, wardens, and administrators.

The application incorporates complaint registration, rule-based automatic staff assignment, severity-based Service Level Agreement (SLA) monitoring, overdue complaint escalation, and WebSocket-based real-time status updates. It also supports role-based access control and student verification or reopening of resolved complaints.

Functional testing was conducted across nine test cases covering authentication, complaint management, staff assignment, SLA monitoring, escalation, real-time updates, access control, and resolution verification. All nine test cases passed during the reported evaluation, indicating that the tested functionalities operated as expected.

Overall, the project provides a structured approach to hostel complaint management by integrating assignment, monitoring, communication, and verification into a single application. The system establishes a foundation for improving transparency and accountability in hostel maintenance operations. Further evaluation under larger workloads and additional operational scenarios can help assess its scalability and reliability.

16. FUTURE SCOPE

The Smart Hostel Complaint Management and Real-Time Resolution System can be extended with additional features to improve accessibility, operational efficiency, and complaint analysis.

1. **Mobile Application:** Develop an Android or iOS application to enable students and maintenance staff to access complaint services conveniently.
2. **Advanced Analytics Dashboard:** Introduce dashboards to analyze complaint categories, resolution durations, staff workloads, and recurring maintenance issues.
3. **Email and SMS Notifications:** Integrate email and SMS services to notify users about complaint assignments, status changes, approaching deadlines, and escalations.
4. **Cloud Deployment:** Deploy the application on a cloud platform to improve accessibility and support centralized management across multiple hostels.
5. **AI-Assisted Complaint Classification:** Explore machine-learning techniques to categorize complaints and recommend priority levels based on complaint descriptions.
6. **Predictive Maintenance:** Analyze historical complaint records to identify recurring infrastructure problems and support preventive maintenance planning.
7. **Performance and Scalability Improvements:** Conduct load testing and optimize the application to support larger numbers of users and concurrent complaint updates.

These enhancements could extend the system beyond its current implementation and provide additional capabilities for hostel administration and maintenance management.

17. REFERENCES

Research Papers and Publications

1. Wong, P. L., and Mahdin, H. (2022). *Design and Development of Online Hostel Management System to Improve Application and Management Process*. Applied Information Technology and Computer Science, 3(2), 356–369. Available at: https://periodical.uthm.edu.my/index.php/aitcs/article/view/7496
2. Alifat, F. A., et al. (2022). *Web-based Campus Complaint Management System (WCCMS)*. International Journal of Computer Trends and Technology, 70(10), 28–36. Available at: https://ijcttjournal.org/archives/ijctt-v70i10p105
3. Iden, J., and Eikebrokk, T. R. (2013). Implementing IT Service Management: A Systematic Literature Review. *International Journal of Information Management*. Available at: https://doi.org/10.1016/j.ijinfomgt.2013.01.004
4. Serrano, J., et al. (2021). An IT Service Management Literature Review: Challenges, Benefits, Opportunities and Implementation Practices. *Information, 12*(3), 111. Available at: https://doi.org/10.3390/info12030111
5. Swain, A. K., and Garza, V. R. (2023). Key Factors in Achieving Service Level Agreements (SLA) for Information Technology (IT) Incident Resolution. *Information Systems Frontiers*. Available at: https://doi.org/10.1007/s10796-022-10266-5
6. Lee, D.-H., et al. (2018). Competency Adjustment and Workload Balancing Framework in Multirobot Task Allocation. *International Journal of Advanced Robotic Systems*. Available at: https://doi.org/10.1177/1729881418812960
7. Gnanagurusubbiah, S., Manivannan, D., and Kumar, A. S. (2026). *StudentConnect: A Digital Solution for Managing Late Permissions and Maintenance Complaints in Hostels*. Premier Journal of Science. Available at: https://premierscience.com/pjs-25-1191/
8. *Automation Tool for Grievances Using Artificial Intelligence*. (2026). *Procedia Computer Science*. Available at: https://doi.org/10.1016/j.procs.2026.06.346

Existing Systems and Institutional Portals

9. Vellore Institute of Technology (VIT). *Hostel Facilities and Administration*. Available at: https://vit.ac.in/campuslife/hostels
10. Vellore Institute of Technology (VIT). *Hostel Wi-Fi Help Desk*. Available at: https://hostelwifi.vit.ac.in/
11. Vellore Institute of Technology, Chennai. *V-HELP Campus Support System*. Available at: https://vhelp.vit.ac.in/vitcc-help-center/
12. National Institute of Technology Rourkela. *Hall Management System*. Available at: https://cwo.nitrkl.ac.in/login
13. Indian Institute of Technology Kanpur. *Campus Care*. Available at: https://www.iitk.ac.in/doip/campus-care

14. National Institute of Technology Patna. *Smart Hostel Complaint Management System*. Available at: https://github.com/nksharma132/smart-hostel-complaint-management-system

15. Government of Bihar. *Hostel Student Support Portal*. Available at: https://sahyog.bihar.gov.in/vidyarathi/HSS/Student_Reg_1.aspx

Technical Documentation

16. Oracle. *Java Documentation*. Available at: https://docs.oracle.com/en/java/

17. Spring. *Spring Boot Reference Documentation*. Available at: https://docs.spring.io/spring-boot/

18. Spring. *Spring Security Reference Documentation*. Available at: https://docs.spring.io/spring-security/reference/

19. Oracle. *MySQL Reference Manual*. Available at: https://dev.mysql.com/doc/

20. Mozilla Developer Network (MDN). *WebSocket API*. Available at: https://developer.mozilla.org/en-US/docs/Web/API/WebSocket

21. Apache Software Foundation. *ActiveMQ Artemis Documentation*. Available at: https://activemq.apache.org/components/artemis/documentation/

22. Apache Software Foundation. *Apache Maven Documentation*. Available at: https://maven.apache.org/guides/

23. Git. *Git Documentation*. Available at: https://git-scm.com/doc

24. GitHub. *GitHub Documentation*. Available at: https://docs.github.com/

Project Source Code

25. Rithish K. *Smart Hostel Complaint Management and Real-Time Resolution System*. Project source code and implementation.

    GitHub repository: https://github.com/rithishk2506/smart-hostel-complaint-system